Worksheet 14 · Standard Form IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Write numbers in and out of standard form • Calculate with numbers in standard form • Solve problems involving very large and very small numbers		
Grade: / 25	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Key notes

Required Form → $a \times 10^n$ where $1 \leq a < 10$

Questions *Standard Form*

Question 1 (2 marks)

(a) Write 1 390 000 in standard form.

.....

(1)

(b) Write 0.005 in standard form.

.....

(1)

Question 2 (3 marks)

(a) Write 8×10^4 as an ordinary number.

.....

(1)

(b) Work out $(3.5 \times 10^5) \div (7 \times 10^8)$
Give your answer in standard form.

.....

(2)

Question 3 (4 marks)

The table shows the volumes, in km³, of four oceans.

Ocean	Volume (km³)
Arctic Ocean	1.88×10^7
Atlantic Ocean	3.10×10^8
Indian Ocean	2.64×10^8
Southern Ocean	7.18×10^7

(a) Write 7.18×10^7 as an ordinary number.

.....

(1)

(b) Calculate the total volume of these four oceans.

..... km³

(2)

The volume of the South China Sea is 9 880 000 km³

(c) Write 9 880 000 in standard form.

.....

(1)

Question 4 (3 marks)

$a = 25 \times 10^{14n}$ where n is an integer.

Find an expression, in terms of n, for $a^{\frac{3}{2}}$
Give your answer in standard form.

.....

Question 5 (2 marks)

(a) Write 2 840 000 000 in standard form.

.....

(1)

(b) Write 2.5×10^{-4} as an ordinary number.

.....

(1)

Question 6 (3 marks)

The table gives the length of the coastline, in kilometres, of each of five oceans.

Ocean	Length of coastline (km)
Arctic	4.539×10^4
Atlantic	1.119×10^5
Pacific	1.357×10^5
Indian	6.653×10^4
Southern	1.797×10^4

(a) Which ocean has the greatest length of coastline?

.....

(1)

(b) Calculate the difference between the length of the Atlantic Ocean's coastline and the length of the Southern Ocean's coastline.
Give your answer in standard form.

..... km

(2)

Question 7 (3 marks)

(b) Work out $\dfrac{9.6 \times 10^{141} + 6.4 \times 10^{140}}{3.2 \times 10^{16}}$

Give your answer in standard form.

.....

(3)

Question 8 (2 marks)

The table gives information about the population, correct to 2 significant figures, of each of five cities in 2018

City	Population (2018)
Ahmedabad	7.7×10^6
Barcelona	5.5×10^6
Chicago	8.8×10^6
Lagos	1.3×10^7
Tokyo	3.7×10^7

(a) Write 8.8×10^6 as an ordinary number.

.....

(1)

(b) Which of these cities had the least population in 2018?

.....

(1)

(c) Work out the difference between the population of Tokyo and the population of Ahmedabad in 2018
Give your answer in standard form correct to 2 significant figures.

Question 9 (3 marks)

(a) Write 6.25×10^{-4} as an ordinary number.

.....

(1)

(b) Work out $(2.4 \times 10^{12}) \div (9.6 \times 10^4)$
Give your answer in standard form.

.....

(2)